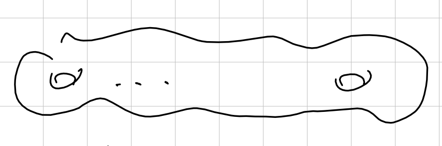
S^n, ~~$S^1\times S^1$~~, $\mathbb{R}P^n$,



$(n\geq 1)$ ↓ $\pi_1=0$.

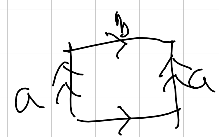
2-dim mfd

$L(p,q)$

$M \# N$. ∧ ∨

$D^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$[0,1]^2$



torus. $\mathbb{Z}\times\mathbb{Z}$

Klein.

Klein. free product.

~~$\mathbb{Z}*\mathbb{Z}$~~

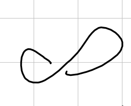
abababb..

$\langle a,b \mid aba=b\rangle$.

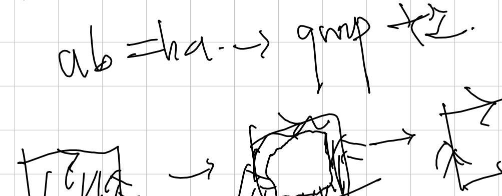
$aba^{-1}b^{-1}=e$.

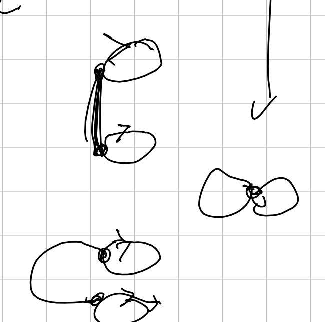
$ab=ba$ → group +Z.

$\langle a,b \mid ab=ba\rangle$ $\mathbb{Z}\times\mathbb{Z}$



$S^1\vee S^1$





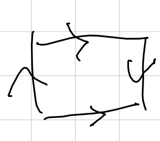
$S^1\vee S^2$

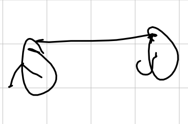
$T: \mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$ / $\mathbb{Z}*\mathbb{Z}$

$T_1: \mathbb{Z}*\mathbb{Z}$

$K_1: \mathbb{Z}*\mathbb{Z}$





S^1 ↕ $\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$ $\mathbb{Z}*\mathbb{Z}$

S^3: 0. ↓ $\mathbb{R}^3$: 0.

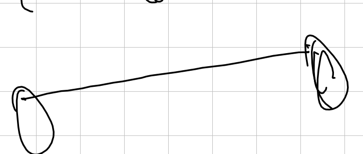
M: n-dim with $(n\geq 3)$

then $\pi_1(M)\simeq\pi_1(M-\bullet)$.

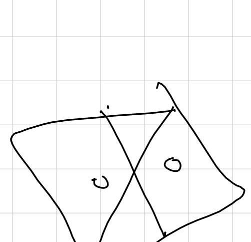
$\simeq\pi_1(M-2pt.)$ $\pi_1(S^n)=0$ $n\geq 2$.

$\pi_1(\mathbb{R}P^n)=\mathbb{Z}_2$ $(n\geq 2)$.

$\simeq\pi_1(M-\{x_1,\ldots\})$.



~~tree~~



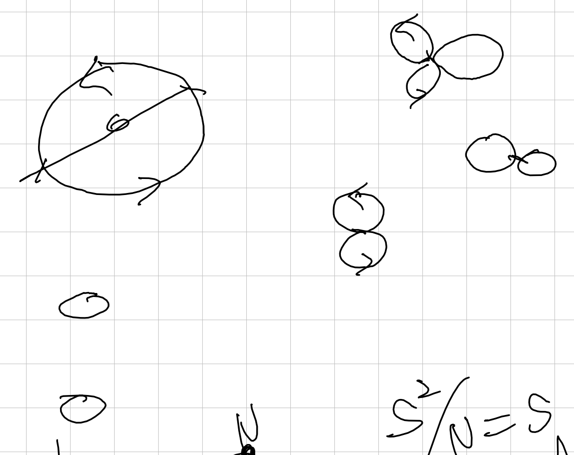
$S^1\vee S^1\vee\mathbb{R}$.

↓

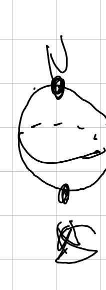
$\mathbb{Z}*\mathbb{Z}$

: $S^1\vee S^1\vee S^1$

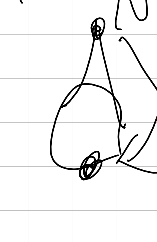
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

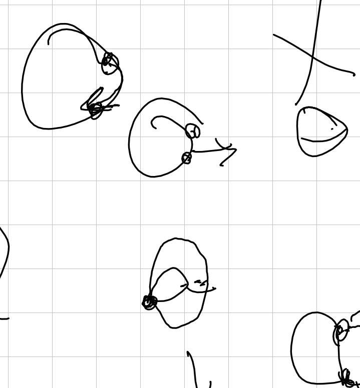


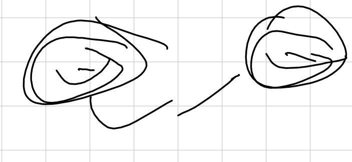
Example.



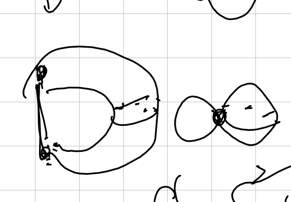
$S^2/N = S$



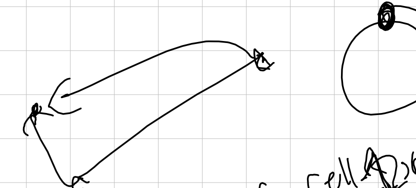




→ $\pi_1(\ldots)$

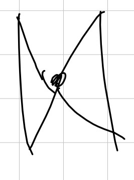


$S^1\vee S^2$



S^1: 0-cell 1개 1-cell 1개.

S^2: 0-cell 1개 2-cell 1개



$S^1\vee S^1$

$\mathbb{R}P^2$: 0-cell: 1개 1-cell: 1개 2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\langle aba^{-1}b^{-1}\rangle$

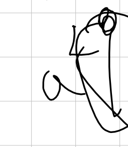
$\mathbb{R}P^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$

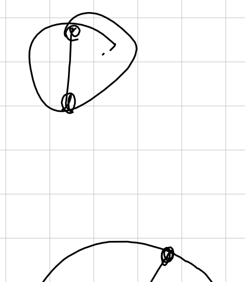
$a^2=1$

~~$\mathbb{R}P^3$: 0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

~~$\pi_1(\mathbb{R}P^3)$~~

$\pi_1(\mathbb{R}P^3)=\pi_1(\mathbb{R}P^2)=\mathbb{Z}_2$.

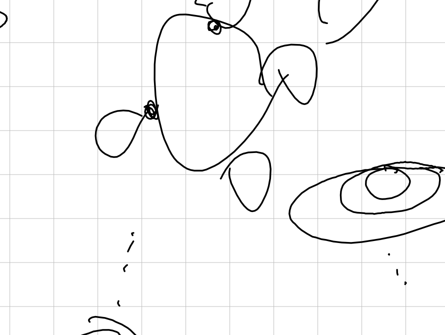


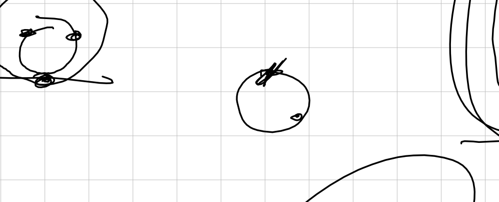


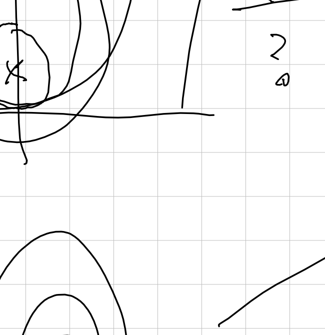
← $x^2+y^2+z^2=1$ ∪ $x^2+y^2\leq 1$ ∪ $(0,0,z)$ $-1\leq z\leq 1$.

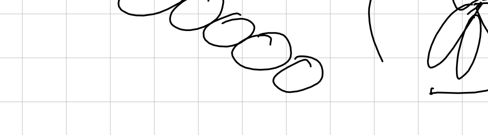
$(\frac{1}{2},\frac{1}{2},\ldots)$

$\mathbb{R}/\mathbb{Z}$









$z=x^4+y^4$

↓ exp

$K_1=f(y,z)$.. $K_2=f_y$..

$\rho=0$

$\frac{1}{\rho_n+z}$.

$\int\frac{\ldots dz}{4}$ $\int\frac{1}{z}dz$ / $2\pi i$

$2\pi i\chi(\Gamma)$ / 0